Contextual Integrity and Age: Rethinking Transmission Principles Across Age Groups

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Abstract

Contextual Integrity (CI) is now a common framework for thinking about privacy. CI defines privacy as appropriate information flow based on contextual norms. Many studies apply CI to digital platforms. However, these studies often assume that norms work the same way for all age groups. Researchers usually treat age as a simple demographic trait of the user. They do not treat age as something that shapes the norms themselves.

This paper argues that age does more than describe a data subject. Age shapes how people interpret transmission principles. Different life stages affect how users understand visibility, audience size, data persistence, and secondary use. A flow of information may follow general contextual rules. Yet that same flow may conflict with the expectations of adolescents or other age groups. If researchers do not model age differences clearly, CI analysis may miss these variations in normative expectations.

This project develops an age-sensitive reading of transmission principles. The paper does not add a new parameter to CI. Instead, it rethinks how we judge normative legitimacy in relation to life-stage expectations. The goal is to ask whether age should be treated as part of CI's core normative logic and what this means for platform governance and regulation.

CCS Concepts

• **Security and privacy → Social aspects of security and privacy**.

Keywords

Contextual Integrity, age differentiation, transmission principles, privacy theory

1 Introduction

1.1 Contextual Integrity and Its Extensions

Contextual Integrity (CI) defines privacy as the appropriateness of information flows within specific social contexts [Nissenbaum 2004, 2009]. The framework evaluates flows through key parameters, including sender, recipient, subject, attribute, and transmission principle. This structure allows researchers to assess whether a data practice aligns with contextual norms [Barth et al. 2006]. CI has been widely used to analyze digital platforms, algorithmic decision systems, and regulatory approaches to data governance [Benthall et al. 2017; Viljoen 2021].

Scholars have extended CI in several important ways. Some researchers have translated CI into formal and computational models that can evaluate information flows in technical systems [Barth et al. 2006; Benthall et al. 2017]. Other scholars have applied CI to questions of data protection law and institutional governance, showing how contextual norms can guide platform design and democratic accountability [Viljoen 2021]. These developments show that CI remains adaptable across domains and continues to serve as a strong theoretical framework for privacy analysis.

1.2 The Problem of Normative Homogeneity

Many extensions of CI still assume that contextual norms apply in the same way to all users within a shared institutional setting. Most analyses treat age as a feature of the subject parameter. Scholars often use age to distinguish minors from adults for legal reasons, but they do not adjust how transmission principles are interpreted. This approach assumes that users who share the same platform also share the same normative expectations.

However, digital platforms bring together very different groups under one technical system. Adolescents and adults use the same infrastructure, yet they may hold different expectations about visibility, persistence, and secondary use. A shared architecture does not guarantee shared norms. When platforms apply uniform rules to diverse life stages, they may overlook important differences in how users interpret information flows.

Research on youth and networked publics shows that privacy expectations change across life stages [Boyd 2014; Livingstone and Helsper 2008]. Adolescents often operate within peer-focused visibility environments shaped by identity development and social judgment. Adults more often use platforms for work, coordination, or instrumental goals. Features such as algorithmic amplification and long-term data storage can disrupt expectations in ways that differ by life stage [Susser et al. 2019].

1.3 Age as a Structuring Condition

This paper argues that age should not be treated only as a demographic feature of the subject. Age should instead be understood as a structuring condition that shapes how transmission principles gain normative meaning. The same information flow can have the same sender, recipient, attribute, and stated transmission condition. Yet that flow may fit general contextual norms and still violate age-specific expectations. Normative meaning does not stay fixed across life stages.

This project does not add a sixth parameter to CI. It changes how the evaluative baseline is defined. Age shapes how users interpret audience scope, data persistence, and acceptable reuse. Norms are filtered through life-stage experience. If analysis ignores life-stage differences, CI assessments may hide real variation in expectations across platform populations.

The need for this shift is clear in today's platforms. Digital systems place adolescents and adults inside the same technical structure. Platforms apply the same default transmission settings to all users. But expectations about visibility and long-term data storage often differ by age. Normative divergence becomes built into the system itself, not just a side effect.

The paper first clarifies how age fits within CI's five parameters. It then models age-differentiated information flows. It finally proposes an age-sensitive refinement that keeps CI's structure intact while improving its normative precision. This approach treats life stage as internal to contextual integrity rather than as an external policy layer.

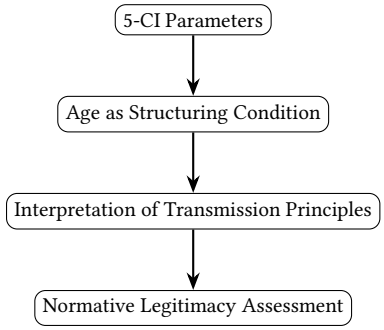


Figure 1: Age as a contextual modifier of transmission principles in CI.

2 Contextual Integrity and Age

2.1 The Structure of Contextual Integrity

Contextual Integrity evaluates privacy by asking whether information flows match context-specific norms [Nissenbaum 2004]. The framework defines these norms through five parameters: sender, recipient, subject, attribute, and transmission principle [Nissenbaum 2009]. An information flow is legitimate when it fits shared expectations within a given social setting.

Transmission principles play a key role in this evaluation. They describe the conditions under which information may properly move from one party to another, such as confidentiality, consent, or functional necessity. Changes in transmission principles often signal privacy problems [Barth et al. 2006].

In many digital applications of CI, age appears only as part of the subject parameter. Laws and policies may distinguish minors from adults. However, analysts often treat transmission principles as fixed across age groups. Expectations about visibility, data persistence, and redistribution are usually assumed to remain stable within the same institutional environment.

2.2 Age as a Structuring Condition

This approach reduces the importance of life stage in shaping norms. Age affects the expectations that define a context. Adolescents and adults hold different social roles, and these roles shape how they think about audience limits and later data use [Boyd 2014].

Adolescents often share information within peer groups and expect limited redistribution. Platform algorithms may expand the audience beyond that group. This expansion can disrupt the expected transmission principle, even when the basic structure of the information flow stays the same [Susser et al. 2019]. Adult users in professional settings may expect wider circulation, but only within role-based boundaries.

Age therefore works as a structuring condition that shapes how transmission principles gain meaning. This argument does not add a sixth parameter to CI. Instead, it clarifies that life stage influences how norms are interpreted. The structure of contextual integrity remains stable, but the baseline for evaluation becomes socially differentiated.

3 Modeling Age-Differentiated Information Flows

This section shows how age affects normative evaluation within Contextual Integrity. It compares information flows that share the same structure but occur at different life stages. The focus is on conceptual comparison, not on empirical testing or data measurement.

3.1 Adolescent Peer Visibility Flow

- **Sender:** Adolescent user
- **Recipient:** Platform
- **Secondary Recipient:** Algorithmically expanded audience
- **Attribute:** Personal expression
- **Expected Transmission Principle:** Peer-bounded visibility

In adolescent peer groups, people usually share information with the expectation that it will stay within a limited circle [Boyd 2014]. They expect visibility to remain within known social boundaries, and they expect little redistribution. Platform algorithms can expand that audience beyond what users anticipate. This expansion can change the meaning of the transmission principle, even if the user has given formal consent. The structure of the information flow stays the same. The change occurs in the normative baseline used to judge that flow.

This difference shows that contextual expectations do not depend only on contracts or platform policies. Social understandings shape expectations about audience, purpose, and relational limits. Algorithmic systems can shift audience scope without adjusting these shared understandings. This shift can create contextual misalignment, even when no explicit rule is broken. This problem becomes especially visible during adolescence, when identity formation and peer recognition give disclosure strong social meaning.

3.2 Adult Professional Data Flow

- **Sender:** Adult user
- **Recipient:** Platform
- **Secondary Recipient:** Advertisers or data brokers
- **Attribute:** Behavioral data
- **Expected Transmission Principle:** Service provision or consent

In adult settings, normative tension often comes from changes in purpose rather than simple audience expansion [Viljoen 2021]. Adult users may expect data processing within clear functional or contractual limits. However, large-scale aggregation, cross-context profiling, and resale in data markets can go beyond those expectations.

The source of concern differs from the adolescent case. Adolescents focus on audience limits in peer spaces. Adults often focus on how their data are reused and commercialized. A service may collect data for one purpose and later use it for profit or strategic influence. The structural parameters of the flow stay the same, but the standard used to judge that flow changes.

These comparisons show that identical parameter structures do not ensure the same normative outcome. Age does not change the formal setup of sender, recipient, or attribute. Age shapes how transmission principles are understood and where vulnerability appears within the flow.

4 Extending Contextual Integrity

Recognizing age as a structuring condition changes how Contextual Integrity is applied. The earlier models show that differences across age groups appear at the level of norm interpretation, not at the level of structural parameters. The five-parameter framework remains stable [Nissenbaum 2009]. However, evaluators must consider that shared digital spaces contain different normative baselines.

Age influences how people understand transmission principles such as confidentiality, consent, and functional necessity. If analysts rely only on adult-centered expectations, they may overlook vulnerabilities that affect specific life stages. This approach therefore calls attention to differences within the same platform context. At the same time, it invites several possible objections.

4.1 Potential Objections

One objection claims that age already appears in the subject parameter of CI, so no further change is needed. This view argues that laws already separate minors and adults, and that this separation captures age-related differences.

However, defining age only as a subject attribute does not explain how transmission principles are understood. Legal rules may require different compliance steps, but they do not clarify how expectations about visibility, persistence, audience scope, or redistribution shift across life stages. The proposal here focuses on how norms are interpreted, not on adding new parameters.

Another objection suggests that contextual norms are flexible and can adjust to age differences without formal revision. This claim may hold in theory. In practice, digital platforms often apply the same technical defaults to diverse users. When systems enforce uniform settings, age-based differences in expectations may remain hidden unless analysts model them directly.

4.2 Normative Implications

Contextual Integrity defines privacy through norms that depend on context. If this is true, then differences within a context create a real theoretical problem, not a minor detail. An age-sensitive approach asks a more specific question. It asks whether an information flow fits the norms that apply to a particular life stage within that context. Analysts cannot assume legitimacy based only on institutional boundaries. They must evaluate flows against different normative baselines that reflect age-based expectations.

This view challenges the idea that digital contexts operate as fully coherent domains. Platform systems often gather diverse groups into the same technical space. These groups hold different expectations about visibility, persistence, and secondary use. When analysts treat the context as uniform, they hide these internal differences. This treatment can obscure vulnerabilities that affect certain life stages more than others.

Recognizing age as a structuring condition strengthens CI's internal logic. This proposal does not add new parameters. It refines how existing norms are interpreted. The framework remains structurally stable, but it becomes more sensitive to differences that exist within shared sociotechnical systems.

4.3 Implications for Platform Governance

Age changes how people interpret transmission principles. So governance must address life-stage differences at the level of system design. Compliance labels alone are not enough. Age-sensitive governance would do more than separate minors and adults through access limits. It would examine whether platform design matches the expectations that apply to different life stages.

Algorithmic amplification systems provide a clear example. Many platforms default to engagement maximization. These defaults can disrupt adolescent expectations of peer-bounded sharing. Teens often expect information to stay within a limited social circle. Amplification systems can expand that audience without changing the formal structure of the flow. Persistent storage and cross-context aggregation raise similar concerns. Downstream commercialization can create risks that vary by life stage, even when users give formal consent. An age-sensitive CI approach shifts attention from demographic labels to the fit between technical design and differentiated normative baselines.

This shift also changes how analysts evaluate system defaults. Many infrastructures reflect adult-centered assumptions about publicity, durability, and reuse. An age-sensitive framework asks whether those assumptions hold across heterogeneous users. It treats age as an analytical lens. It examines how well technical systems align with socially situated expectations.

Regulation also faces limits when it relies only on age thresholds. Legal categories trigger compliance duties, but they do not capture the full range of normative variation in digital contexts. A CI-informed regulatory approach could treat age as a structuring dimension when assessing embedded transmission principles. This move would redirect attention from demographic classification to normative alignment.

Integrating age into CI strengthens the framework's practical relevance. It improves the ability to evaluate whether sociotechnical systems respect differentiated expectations across diverse digital publics. Future research can explore how age-differentiated transmission principles might be formalized in evaluative tools or embedded in platform auditing practices.

5 Conclusion

This paper argues that age should not be treated only as a feature of the data subject in Contextual Integrity. Age should be understood as a structuring condition that shapes how transmission principles are interpreted. The analysis models information flows across different life stages. The comparison shows that normative legitimacy can shift across age groups, even when the five structural parameters stay the same.

This intervention does not add a new parameter to CI. The five-part structure remains stable. The change occurs at the level of normative baseline formation. Age shapes expectations about visibility, audience scope, persistence, and secondary use. These expectations influence how transmission principles gain meaning in a specific context. Life-stage differentiation works as an internal modifier of contextual norms, not as an external regulatory layer.

Digital platforms now operate as multi-generational systems. They apply uniform transmission defaults to users with different expectations. Adult-centered baselines often guide evaluation. This approach can hide life-stage–specific vulnerabilities that exist within shared infrastructures. Integrating age into CI's normative logic improves the framework's ability to assess complex sociotechnical systems while keeping its formal structure intact.

Highlighting age does not expand CI beyond its original design. It shows that contextual norms are already internally differentiated. Recognizing this differentiation opens space for further work on how CI can address heterogeneous digital publics and support more responsive platform governance and regulatory design.

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